



Math 101: Calculus I

Second Exam, 11 December 2017

Duration: 90 minutes

- In question 1, select the correct answer (no partial credit is given).
- In questions 2-6, show all your work.
- Calculators and communication devices are not allowed in the examination room.

Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (20 pts)	
Q2 (10 pts)	
Q3 (10 pts)	
Q4 (10 pts)	
Q5 (10 pts)	
Q6 (40 pts)	
Total	

1. [$4 \times 5 = 20$ pts.] In each of the following multiple choice questions, select the correct answer.

I. Let $f(x) = \log_{10}(x^2 + 1)$. Then:

(A) $f'(x) = \frac{2x}{x^2 + 1}$.

(B) $f'(x) = \frac{2x}{(x^2 + 1) \ln 10}$.

(C) $f'(x) = \frac{1}{x^2 + 1}$.

(D) $f'(x) = \frac{1}{(x^2 + 1) \ln 10}$.

(E) None of the above.

II. Let $f(x) = x^2 + 1$. On the interval $[-1, 1]$:

(A) 1 is the absolute minimum value of f and 2 is the absolute maximum value of f .

(B) $f(0.5)$ is a local minimum value of f .

(C) f has no absolute extreme values.

(D) $f(0.5)$ is a local maximum value of f .

(E) None of the above.

III. $\lim_{x \rightarrow \infty} \frac{\sinh(x)}{8e^x} =$

(A) ∞ .

(B) $\frac{1}{4}$.

(C) 0.

(D) $\frac{1}{16}$.

(E) None of the above.

IV. Let $f(x) = \ln(x^2 - 1)$. Then:

(A) $x = 1$ is a critical number of f .

(B) $x = -1$ is a critical number of f .

(C) $x = 0$ is a critical number of f .

(D) $x = -2$ is a critical number of f .

(E) None of the above.

2. [10 pts.] Find the slope of the tangent line at the point $(0, 0)$ to the curve with equation

$$e^{xy} + \sin(x^2) = \cosh(x) + y.$$

3. [10 pts.] The height of a triangle is increasing at a rate of 1 cm/min while the area of the triangle is increasing at a rate of 2 cm²/min. At what rate is the base of the triangle changing when the height is 10 cm and the area is 100 cm²?
4. [10 pts.] Use linear approximation to estimate $f(2.1)$, where $f(x) = \sqrt{x^3 + 1}$.
5. [10 pts.] Given a function $f(x)$ such that $f(6) = 5$ and $f'(x) \leq -5$ for all $x \in [6, 10]$. How large can $f(10)$ possibly be?

6. [$4 \times 10 = 40$ pts.] Let $f(x) = \frac{e^x}{1 - e^{x-1}}$.

(a) [$5 + 5 = 10$ pts.] Find the vertical and the horizontal asymptotes of the graph of f , if any.

(b) [$5 + 2.5 + 2.5 = 10$ pts.]

i. Show that $f'(x) = \frac{e^x}{(1 - e^{x-1})^2}$.

ii. Find the intervals on which f is increasing and the intervals on which f is decreasing, if any.

iii. Find the local maximum and minimum values of f , if any.

(c) [$5 + 5 = 10$ pts.] Given that $f''(x) = \frac{e^x + e^{2x-1}}{(1 - e^{x-1})^3}$.

i. Find the intervals on which the graph of f is concave upward and the intervals on which the graph of f is concave downward, if any.

ii. Find the points of inflection, if any.

(d) [10 pts.] Sketch the graph of f .

1. [$4 \times 5 = 20$ pts.] In each of the following multiple choice questions, select the correct answer.

I. Let $f(x) = \log_{10}(x^2 + 1)$. Then:

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(E) None of the above.

II. Let $f(x) = x^2 + 1$. On the interval $[-1, 1]$:

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2. [10 pts.] Find the slope of the tangent line at the point $(0, 0)$ to the curve with equation

$$e^{xy} + \sin(x^2) = \cosh(x) + y.$$

Differentiating implicitly wrt x we get: $e^{xy}(y + xy') + 2x \cos(x^2) = \sinh(x) + y'$.

At the point $(0, 0)$ $y' = 0$.

3. [10 pts.] The height of a triangle is increasing at a rate of 1 cm/min while the area of the triangle is increasing at a rate of 2 cm²/min. At what rate is the base of the triangle changing when the height is 10 cm and the area is 100 cm²?

Let A , b and h be the area, base and height of the triangle at time t .

We are given that $h' = 1$ cm/min and $A' = 2$ cm²/min. We need to find b' when $A = 100$ cm² and $h = 10$ cm.

We have $A = \frac{1}{2}(bh)$. Differentiating wrt time t we get: $A' = \frac{1}{2}(b'h + h'b)$.

Therefore, $b' = \frac{2A' - h'b}{h}$.

When $A = 100$ and $h = 10$ we get $b = \frac{2A}{h} = 2(100)/10 = 20$ cm.

Thus, $b' = \frac{2A' - h'b}{h} = (4 - 20)/10 = -1.6$ cm/min.

The base of the rectangle is decreasing at a rate of 1.6 cm/min.

4. [10 pts.] Use linear approximation to estimate $f(2.1)$, where $f(x) = \sqrt{x^3 + 1}$.

$$f(x) \approx f(2) + f'(2)(x - 2) \text{ when } x \text{ is near } 2.$$

We have $f(2) = 3$, $f'(x) = \frac{3x^2}{2\sqrt{x^3 + 1}}$ and $f'(2) = 2$.

Thus, $f(2.1) \approx 3 + 2(2.1 - 2) = 3.2$.

5. [10 pts.] Given a function $f(x)$ such that $f(6) = 5$ and $f'(x) \leq -5$ for all $x \in [6, 10]$. How large can $f(10)$ possibly be?

We are given that $f'(x) \leq -5$ for all $6 \leq x \leq 10$, so certainly f is continuous on $[6, 10]$ and differentiable on $(6, 10)$. Therefore, using the MVT on $[6, 10]$, there exists a number $c \in (6, 10)$ such that $f(10) - f(6) = (10 - 6)f'(c)$ i.e., $f(10) = f(6) + 4f'(c) = 5 + 4f'(c)$.

Now since $f'(c) \leq -5$, $f(10) = 5 + 4f'(c) \leq 5 + 4(-5) = -15$.

Hence the largest possible value of $f(10)$ is -15 .

6. $[4 \times 10 = 40 \text{ pts.}]$ Let $f(x) = \frac{e^x}{1 - e^{x-1}}$.

$$D_f = (-\infty, 1) \cup (1, \infty).$$

(a) We have: $\lim_{x \rightarrow \infty} \frac{e^x}{1 - e^{x-1}} = -e$ and $\lim_{x \rightarrow -\infty} \frac{e^x}{1 - e^{x-1}} = 0$.

Therefore, the lines $y = -e$ and $y = 0$ are horizontal asymptotes of the graph of f .

Furthermore, $\lim_{x \rightarrow 1^\pm} \frac{e^x}{1 - e^{x-1}} = \mp \infty$ and hence $x = 1$ is a vertical asymptote of the graph of f .

(b)
$$f'(x) = \frac{(e^x)(1 - e^{x-1}) - (e^x)(-e^{x-1})}{(1 - e^{x-1})^2} = \frac{e^x}{(1 - e^{x-1})^2}.$$

Since $f'(x) = \frac{e^x}{(1 - e^{x-1})^2}$, $f'(x) > 0$ for all $x \neq 1$.

Thus f is increasing on its domain. Furthermore, f has no critical numbers. Hence f has no local extreme values.

(c) Since
$$f''(x) = \frac{e^x + e^{2x-1}}{(1 - e^{x-1})^3}, \quad f''(x) > 0 \text{ on } (-\infty, 1) \text{ and } f''(x) < 0 \text{ on } (1, \infty).$$

Therefore, the graph of f is concave upward on $(-\infty, 1)$ and it is concave downward on $(1, \infty)$.

There are no inflection points (f is discontinuous at $x = 1$).

(d) Sketch

